

AAanalysis

v1.1 Cheat Sheet

Sequence → physicochemical scales → interpretable features → explainable ML → biological mechanism

[pip install aaanalysis](#) · [aaanalysis.readthedocs.io](#)

AAanalysis is a Python framework for interpretable, sequence-based protein prediction. It turns sequences into physicochemical features (CPP), trains explainable models, and traces every prediction back to a residue × property × group comparison — robust for small datasets.

Install · Import · Load

core + [pro]

```
# Python >= 3.11
pip install aaanalysis          # core
pip install 'aaanalysis[pro]'  # SHAP, FIMO, Bio

import numpy as np
import matplotlib.pyplot as plt
import aaanalysis as aa
df_seq = aa.load_dataset(name='DOM_GSEC') # γ-secretase
labels = df_seq['label'].to_list()
df_scales = aa.load_scales()
```

The Golden Workflow

canonical pipeline

- LOAD** load_dataset · load_scales → **df_seq** · **df_scales**
- PARTS** SequenceFeature.get_df_parts → **df_parts**
- FEATURES** Part × Split × Scale · CPP.run → **df_feat**
- MODEL** TreeModel.fit · dPULearn.fit → **feat_importance** · **labels_**
- EXPLAIN** CPPPlot.feature_map · ShapModel → **figure** · **feat_impact**

Prediction Task Levels

task → setup

Residue AA_*
unit: sliding window (aa_window_size)
ref: non-site windows / shuffled background

positional

Domain DOM_*
unit: part-set jmd_n · tmd · jmd_c (from tmd_start / tmd_stop)
ref: labelled A vs B groups

both

Protein SEQ_*
unit: whole chain (composition)
ref: labelled groups / composition-matched background

compositional

Sequence Anatomy

the TMD model

TMD Target Middle Domain — the central segment of interest (e.g. transmembrane stretch, binding region).

JMD Juxta Middle Domain — the flanks adjoining the TMD (jmd_n on the N-side, jmd_c on the C-side).

JMD-N

TMD

JMD-C

0 tmd_start tmd_stop len(seq)
JMD widths set globally: aa.options['jmd_n_len'] · ['jmd_c_len'].

1: positives 0: rel-negatives 2: unlabeled

Feature Ontology

Part × Split × Scale

PART where on the sequence

tmd · jmd_n · jmd_c · tmd_jmd · jmd_n_tmd_n · tmd_c_jmd_c

SPLIT how to read the part

Segment — contiguous · Pattern — sparse pairs · PeriodicPattern — i, i+3/4

SCALE which physicochemical property

AAontology (~600 scales) · hydrophobicity · charge · helix propensity · shape

TMD × Segment × hydrophobicity → membrane insertion

JMD × Pattern × net charge → electrostatic recognition

TMD × PeriodicPattern × helix → α-helical interface

CPP Strategies

via split_kws

Compositional ≈ sequence/protein-level

one whole-part average (composition-like, position-agnostic)

```
split_kws = sf.get_split_kws(
    split_types="Segment",
    n_split_min=1, n_split_max=1)
cpp = aa.CPP(df_parts=df_parts,
             split_kws=split_kws)
```

Positional ≈ residue-/region-level

sub-segments and/or patterns resolved to positions

```
split_kws = sf.get_split_kws(
    split_types=["Segment", "Pattern",
                 "PeriodicPattern"],
    n_split_max=5, steps_pattern=[3, 4],
    steps_periodicpattern=[3, 4])
cpp = aa.CPP(df_parts=df_parts,
             split_kws=split_kws)
```

Domain level uses both. → CPP strategies: see the CPP tutorial (docs).

Which Module Should I Use?

intent → module

Explore sequence patterns / composition →

AAlogo

Discover discriminative physicochemical features →

CPP

Reduce redundant amino acid scales →

AAclust

Train with positives + unlabeled data →

dPULearn

Train an interpretable classifier →

TreeModel

Explain a prediction (per feature / sample) →

ShapModel [pro]

Visualize CPP features →

CPPPlot

Data & Preparation

load · encode · clean

Load benchmark sequences

load_dataset(name) → df_seq

Load AAontology scales

load_scales() → df_scales

Load precomputed features

load_features(name) → df_feat

Read / write FASTA

read_fasta(file) → df_seq

Encode sequences (one-hot / int)

SequencePreprocessor().encode_*(seqs)

Cluster redundant homologs

filter_seq(df_seq) [pro]

Feature Engineering

parts · CPP · scales

SequenceFeature → sf

sf = aa.SequenceFeature()

· split sequence into parts

sf.get_df_parts(df_seq) → df_parts

· assemble feature matrix X

sf.feature_matrix(df_feat, df_parts) → X

Discover discriminative features

CPP(df_parts).run(labels) → df_feat *

Sweep CPP configs (grid)

CPPGrid().run(...) · .eval() → ranked configs

Simplify → interpretable scales

CPP.simplify(df_feat, labels)

Reduce redundant scales

AAclust().fit(X) [Wrapper]

Drop correlated features

NumericalFeature().filter_correlation(X)

Numerical Feature Sources

v1.1

PLM · structure · PTM

PLM embeddings

EmbeddingPreprocessor().encode(...) → dict_num

Structure / DSSP / PAE

StructurePreprocessor().encode_dssp(...) [pro]

PTM / site annotations

AnnotationPreprocessor().encode(...) [pro]

Combine sources

combine_dict_nums([...]) → dict_num

Slice to parts

NumericalFeature().get_parts(...) → dict_num_parts

Numerical CPP

CPP(df_parts).run_num(dict_num_parts, labels) → df_feat

Modeling & Explainability

Train + RFE + MC importance

TreeModel().fit(X, labels) [Wrapper]

Train with positives + unlabeled

dPULearn().fit(X, labels) [Wrapper]

Per-feature / sample SHAP impact

ShapModel().fit(X, labels) [pro]

Sequence Analysis

logos · motifs

Position-specific logo

AAlogo().get_df_logo(df_parts)

Sample sequence windows

AAWindowSampler().sample(df_seq)

Pairwise sequence similarity

comp_seq_sim(df_seq) [pro]

Scan motifs (FIMO / MEME)

scan_motif(df_seq, pwm) [pro]

Metrics & Plotting

utilities

Adjusted AUC (class imbalance)

comp_auc_adjusted(X, labels)

BIC score · KL divergence

comp_bic_score(X, labels) · comp_kld

Per-protein / detection (v1.1)

comp_per_protein_ap · comp_detection_metrics

Global plot style & fonts

plot_settings(font_scale)

Colours & standalone legend

plot_get_clist(n) · plot_legend(ax)

Protein Design

(to be extended) v1.1

In-silico point mutations

AAMut · AAMutPlot

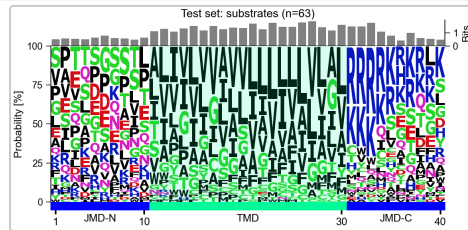
Sequence-design libraries

SeqMut · SeqMutPlot

AAlogo — see the data

dataset at a glance

```
import numpy as np, matplotlib.pyplot as plt, aaanalysis as aa
df_seq = aa.load_dataset(name='DOM_GSEC') # γ-secretase
labels = list(df_seq['label']); df_scales = aa.load_scales()
sf = aa.SequenceFeature()
df_parts = sf.get_df_parts(df_seq=df_seq,
    list_parts=['tmd', 'jmd_n', 'jmd_c'])
aa.plot_settings(font_scale=0.7)
# aal_kws builds df_logo + bits bar for you
aa.AAlogoPlot().single_logo(
    aal_kws=dict(df_parts=df_parts, labels=labels,
        label_test=1, tmd_len=20),
    name_data='Test set: substrates')
plt.tight_layout(); plt.show()
```



CPP — feature

top feature · REF vs TEST

```
# default parts + a redundancy-reduced set of 100 scales
df_parts = sf.get_df_parts(df_seq=df_seq)
sel = aa.AAclust().fit(np.array(df_scales).T,
    names=list(df_scales), n_clusters=100).medoid_names
cpp = aa.CPP(df_parts=df_parts, df_scales=df_scales[sel])
df_feat = cpp.run(labels=labels, n_filter=100)
X = sf.feature_matrix(df_feat['feature'], df_parts)
tm = aa.TreeModel(); tm.fit(X, labels=labels)
df_feat = tm.add_feat_importance(df_feat=df_feat, sort=True)
cpp_plot = aa.CPPPlot(); aa.plot_settings()
# distribution of the top feature (feat_rank=1 of the sorted
df_feat)
cpp_plot.feature(feature=df_feat, feat_rank=1, df_seq=df_seq,
    labels=labels, name_test='substrates', name_ref='non-subs.')
plt.tight_layout(); plt.show()
```

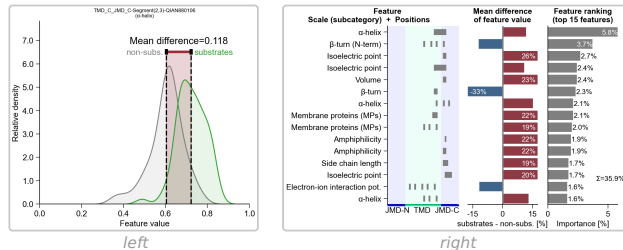
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CPP — ranking

top features · effect + importance

```
# same df_feat — rank the top discriminative features
aa.plot_settings(font_scale=0.6)
cpp_plot.ranking(df_feat=df_feat, n_top=15, rank=True,
    name_test='substrates', name_ref='non-subs.')
plt.tight_layout(); plt.show()
```

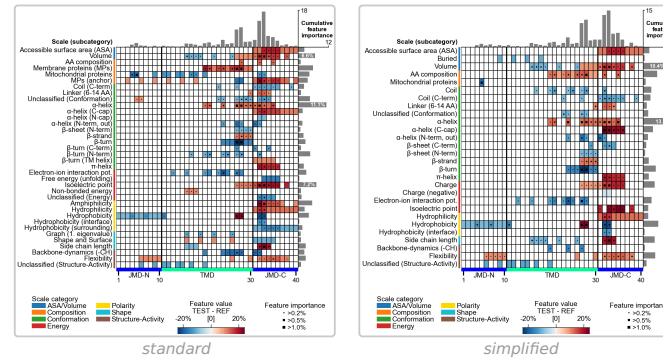
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CPP — feature map

group level · full vs simplified

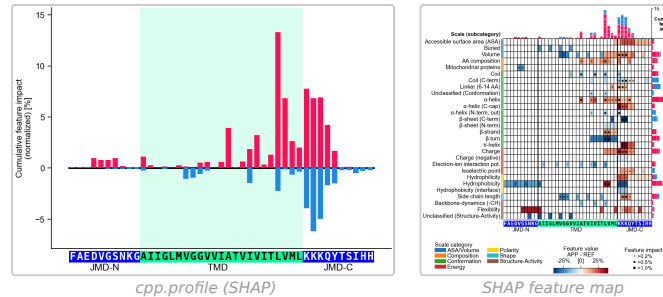
```
# global Part × Split × Scale map — all AAontology scales
cpp_plot = aa.CPPPlot(); aa.plot_settings(font_scale=0.65)
cpp_plot.feature_map(df_feat=df_feat) # left
# CPP.simplify → fewer, interpretable correlated scales
df_feat = cpp.simplify(df_feat=df_feat, labels=labels)
cpp_plot.feature_map(df_feat=df_feat) # right
plt.tight_layout(); plt.show()
```



ShapModel — explain a prediction

sample level · [pro]

```
# fuzzy labeling: APP's label is its soft prediction score (0.6,
not 1)
i = list(df_seq['entry']).index('P05067') # APP
y = [float(v) for v in labels]; y[i] = 0.6
sm = aa.ShapModel(); sm.fit(X, labels=y, fuzzy_labeling=True)
df_feat = sm.add_feat_impact(df_feat=df_feat,
    sample_positions=i, names='APP')
args_seq = {k + ' seq': v for k, v in sf.get_df_parts(
    df_seq=df_seq).loc['P05067'].to_dict().items()}
ka = dict(col_imp='feat_impact APP', shap_plot=True, **args_seq)
cpp_plot.profile(df_feat=df_feat, **ka) # left
cpp_plot.feature_map(df_feat=df_feat, name_test='APP', **ka) #
right
plt.tight_layout(); plt.show()
```



AAclust — clusters

scale reduction · Wrapper

```
X = np.array(df_scales).T
aac = aa.AAclust()
aac.fit(X, names=list(df_scales), n_clusters=10)
aac.medoid_names_ # redundancy-reduced scales

aac_plot = aa.AAclustPlot()
aac_plot.centers(X, labels=aac.labels_)
plt.tight_layout(); plt.show()
```

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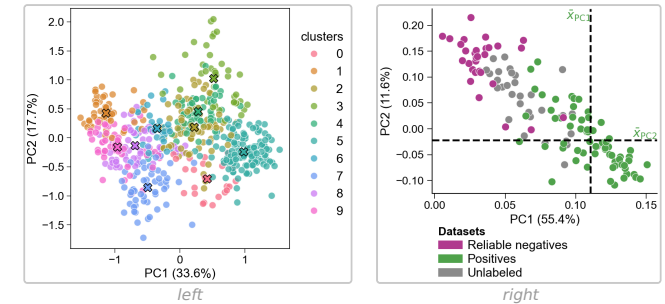
dPUlearn — PCA

reliable negatives · Wrapper

```
# labels: 1 = positive, 2 = unlabeled
n_pos = sum(np.array(labels) == 1)
dpul = aa.dPUlearn()
dpul.fit(X=X, labels=labels, n_unl_to_neg=n_pos)
df_pu = dpul.df_pu_ # 1 pos · 0 rel-neg · 2 unl
```

```
dpul_plot = aa.dPUlearnPlot()
dpul_plot.pca(df_pu=df_pu, labels=labels)
plt.tight_layout(); plt.show()
```

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The 5-Minute Example

copy-paste this

```
import matplotlib.pyplot as plt
import aaanalysis as aa
df_seq = aa.load_dataset(name='DOM_GSEC')
labels = df_seq['label'].to_list()
sf = aa.SequenceFeature()
df_parts = sf.get_df_parts(df_seq=df_seq)
cpp = aa.CPP(df_parts=df_parts)
df_feat = cpp.run(labels=labels)
X = sf.feature_matrix(features=df_feat['feature'],
                      df_parts=df_parts)
tm = aa.TreeModel(); tm.fit(X, labels=labels)
df_feat = tm.add_feat_importance(df_feat=df_feat)
cpp_plot = aa.CPPPlot()
aa.plot_settings()
cpp_plot.feature_map(df_feat=df_feat)
plt.tight_layout(); plt.show()
```

aa.options

system-level settings

```
aa.options['random_state'] = 42
aa.options['verbose'] = True
aa.options['allow_multiprocessing'] = True

# TMD model — JMD flank widths
aa.options['jmd_n_len'] = 10
aa.options['jmd_c_len'] = 10

# plot labels & system-level scales
aa.options['name_tmd'] = 'P5-P5' # e.g. cleavage-site
prediction
aa.options['df_scales'] = my_scales
```

Glossary

- Prediction level** — Residue (AA_*) · Domain (DOM_*) · Protein (SEQ_*) — the unit a task predicts at; sets the dataset, the part, and the reference.
- Part** — Named segment used as feature input: tmd, jmd_n, jmd_c, tmd_jmd, jmd_n_tmd_n, tmd_c_jmd_c.
- Split** — How a scale is read across a part: Segment (contiguous), Pattern (sparse), PeriodicPattern (i, i+3/4).
- Scale** — AA → ℝ mapping. AAontology ships ~600 curated scales in two-level categories.
- Feature** — (Part × Split × Scale) — the atomic, residue-grounded, interpretable unit of CPP.

Class ↔ Plot Class	mirrored API	
SequencePreprocessor	—	
EmbeddingPreprocessor	—	v1.1
StructurePreprocessor [pro]	—	v1.1
AnnotationPreprocessor [pro]	—	v1.1
CPP	CPPPlot	
AAclust	AAclustPlot	Wrapper
AAlogo	AAlogoPlot	
dPULearn	dPULearnPlot	Wrapper
TreeModel	—	Wrapper
ShapModel [pro]	—	Wrapper
AAMut	AAMutPlot	to be extended
SeqMut	SeqMutPlot	to be extended
AAWindowSampler	—	
SequenceFeature	—	
NumericalFeature	—	

How to Cite

if you use AAanalysis

- AAclust**
Breimann & Frishman (2024a), AAclust: k-optimized clustering for selecting redundancy-reduced sets of amino acid scales
Bioinformatics Advances
[Breimann24a]
- AAontology**
Breimann et al. (2024b), AAontology: An ontology of amino acid scales for interpretable machine learning
Journal of Molecular Biology
[Breimann24b]

- CPP & dPULearn**
Breimann & Kamp et al. (2025), Charting γ-secretase substrates by explainable AI
Nature Communications
[Breimann25a]

Design Principles

- Explicit over implicit — DataFrames everywhere
- Wrappers (.fit / .predict / .eval) set trailing *_ attributes after fit
- Biological interpretability is first-class
- Small-data robust and reproducible (layered seeds)

canonical vocabulary · CONTEXT.md

- Compositional vs positional** — How split_kws resolves locality: a whole-part average (compositional) vs sub-region/position-resolved (positional).
- df_seq** — Sequence table: entry, sequence, label, TMD bounds (tmd_start, tmd_stop).
- df_parts** — Wide table — one column per part (tmd, jmd_n, jmd_c, ...).
- df_feat** — Ranked features: feature, abs_auc, mean_dif, p_val, positions, scale, category.
- Wrapper** — sklearn-style class — .fit / .predict / .eval, sets trailing *_ attributes after fit.

- Plot class** — *Plot mirror of an analytical class — same arguments, visualization only.
- PU labels** — dPULearn input: 1 = positive, 2 = unlabeled. Output: 1 / 0 (reliable-negative) / 2.
- CPP** — Comparative Physicochemical Profiling — discovers ranked Part × Split × Scale features.
- AAontology** — Two-level scale taxonomy; CPP uses its categories to organize and rank features.